Porush Yadav

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Education

- **MS, Mathematics and Scientific Computing** — IIT Kanpur CPI: 9.8 / 10 **2021**
- **BS, Mathematics and Scientific Computing** — IIT Kanpur CPI: 7.8 / 10 **2021**

Work Experience

Career Break — UPSC Civil Services Examination Preparation **Dec 2021 – Jun 2026**

- Cleared the Preliminary Examination **thrice** (top 1% of ∼10,00,000 candidates) with Mathematics optional.

Axis Bank Limited — Manager (IT: Business System Analyst), Mumbai **Aug 2021 – Nov 2021**

- Validated an automated OCR and image-recognition pipeline for KYC document classification (PAN, Aadhaar, DL) and structured form field extraction.
- Conducted root-cause analysis on recognition and extraction failures across degraded or non-standard document scans, resolving edge-case anomalies through document pre-processing fixes and vendor model feedback.

Samsung R&D Institute India — Research Intern, Bangalore **May 2020 – Jul 2020**

- Researched 6G IQ data compression techniques (Vector Quantization, bit-vector compression, LPC) to minimize EVM degradation across uplink and downlink channels. Benchmarked compression models against IQ signal properties and evaluated Machine Learning approaches for adaptive uplink/downlink signal compression.

Earlier ML Internships (eJET Systems | EZ Technologies), Noida **Dec 2018 – Jul 2019**

- Built a hotel chatbot with a Keras GRU intent classifier, NLTK entity extractor, and KNN food recommender.
- Built a real-time multi-face attendance system using OpenCV, pre-trained DNNs, and MongoDB.

Projects

PDF Study Assistant (RAG) | Python, LangChain, FAISS, BM25, FastAPI, Gemini/Groq **Oct 2026**

- Architected an end-to-end RAG pipeline for a 193-page report, delivering accurately cited answers and strict out-of-domain refusals using local MiniLM embeddings and FAISS.
- Designed an evaluation harness (hit@k sweep) on gold-labeled queries, achieving 100% factual retrieval; boosted multi-page answer coverage from 42% to 73% by implementing hybrid (dense + BM25) retrieval.
- Shipped a FastAPI chat interface featuring interactive PDF citations, automatic model fallback, rate-limit backoff, and output citation verification.

Consumer Complaint Analytics | Python, pandas, Dask, scikit-learn, Parquet, Streamlit **July 2026**

- Processed an 8.8 GB (15.2M rows) CFPB database within RAM limits using a Dask & Parquet pipeline.
- Trained a TF-IDF & Logistic Regression NLP classifier on 3.3M narratives: 87% accuracy (vs. 60.6% baseline).
- Uncovered a surge in templated filings (to 77%) via Cramér's V and chi-square tests; deployed on Streamlit.

Financial Market Surveillance | Python, pandas, NumPy, scikit-learn, yfinance, Jupyter **June 2026**

- Built an Isolation Forest anomaly-detection pipeline monitoring 5,800+ days of USD–INR exchange rates.
- Mapped 74.3% of flagged anomalies to known crises (2008 GFC, 2020 COVID) using rolling-volatility features.
- Developed a Market Stress Index capturing a 5.09× baseline volatility spike during peak historical stress.

European Option Pricing under Black-Scholes (MS Project) **Aug 2020 – Nov 2020**

- Analyzed risk-neutral valuation and delta-hedging of vanilla and binary European options via Black-Scholes framework, including option Greeks and implied volatility. Simulated volatility surface for MSFT stock in R.

Technical Skills

- **Languages:** Python, C++, C, SQL, R, MATLAB
- **AI & Generative AI:** PyTorch, LangChain, RAG, FAISS, HuggingFace, LLM APIs (Gemini, Groq), BM25
- **Data Science & ML:** NLP, scikit-learn, pandas, NumPy, TensorFlow, Keras, NLTK, Dask, SHAP
- **Frameworks & Tools:** FastAPI, Streamlit, Parquet, Git

Relevant Coursework

- **Computer Science:** Machine Learning, Computational Cognitive Science, Data Structures & Algorithms
- **Statistics:** Time Series Analysis, Stochastic Processes, Statistical Inference, Data Analysis, Sampling Theory
- **Maths:** Financial Mathematics, ANN/ML Approach for Differential Equations, Linear Algebra, Calculus

Certifications

- Machine Learning Specialization; AI Code Review; Applying Machine Learning to your Data with GCP

Achievements & Scholarships

- **NTSE** Scholarship and **INSPIRE** Scholarship for Higher Education (SHE), Government of India